Lesson 10: Data Preparation and Transformation using Tableau Prep

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What you will learn ?
a) Connect and load data into Tableau Prep
b) Clean and profile data
c) Group and standardize values
d) Merge mismatched fields
e) Perform aggregations in Prep
f) Apply filters and transformations
g) Create and automate data preparation flows

You need Tableau Prep Builder for this Lesson. You need to download and install. Here are the steps :

1. Download

- Download the Windows installer (.exe) directly from the official **Tableau Prep Builder Releases Page** *(or use the* ***Tableau Prep Free Trial Page****)*.

2. Install

- Open your **Downloads** folder and run the .exe file.
- Accept the license agreement, click **Install**, and confirm **Yes** on the Windows prompt.

3. Activate & Launch

- Open **Tableau Prep Builder** from the Start Menu.
- Select **Activate with a product key** (enter your Tableau Desktop/Student key) or **Start a Free Trial**.

Dataset: Custom Designed Students Dataset [csv files]

You can also create database using Python code. Here is the code

```
import pandas as pd
import numpy as np

# We need custom CSV datasets specifically designed for Tableau
Prep features in Practical 10:
# a) Connect & load data
# b) Clean and profile data (data type issues, extra spaces,
nulls, outliers)
# c) Group and standardize values (pronunciation/spelling
variations, e.g. "Comp Sci", "CS", "Computer Science", "CSE",
"Electronic Eng", "ECE")
# d) Merge mismatched fields (e.g., across two unioned files
or separate columns: "Maths" vs "Mathematics", "Std_ID" vs
"Student_ID")
# e) Perform aggregations in Prep (e.g., Aggregate step: Total
Score, Subject Average, Group by Student/Term)
# f) Apply filters and transformations (e.g., Filter out
dropped students, calculate weighted GPA / Grade, split
columns)
```

```
# g) Create and automate data preparation flows (Output step
to .hyper / .csv)

# Table 1: Raw Student Assessment Log (Term 1) ->
Student_Assessments_Term1.csv
# Contains messy department names for (c) Group & Standardize,
inconsistent casings, text dates, extra spaces

term1_data = {
    'Student_ID': ['S101', 'S101', 'S102', 'S102', 'S103',
'S103', 'S104', 'S104', 'S105', 'S105', 'S106', 'S106',
'S107', 'S107', 'S108', 'S108'],
    'Student_Name': [' Aarav Sharma ', 'Aarav Sharma', 'priya
VERMA', 'priya VERMA', 'Rohan Gupta', 'Rohan Gupta', 'Sneha
Patel', 'Sneha Patel', 'aditya Rao ', 'aditya Rao ', 'Ananya
Singh', 'Ananya Singh', 'Rahul Nair', 'Rahul Nair', 'Kavya
Reddy', 'Kavya Reddy'],
    'Department': ['Computer Science', 'Comp Sci',
'Electronics', 'ECE', 'CS', 'CSE', 'Information Tech', 'IT',
'Mechanical', 'Mech Engg', 'Computer Science', 'CS', 'ECE',
'Electronics Engg', 'IT', 'Info Tech'],
    'Assessment_Type': ['Midterm', 'Final', 'Midterm',
'Final', 'Midterm', 'Final', 'Midterm', 'Final', 'Midterm',
'Final', 'Midterm', 'Final', 'Midterm', 'Final', 'Midterm',
'Final'],
    'Exam_Date': ['2025/09/15', '2025/11/20', '15-09-2025',
'20-11-2025', '2025.09.15', '2025.11.20', '2025/09/15',
'2025/11/20', '2025/09/15', '2025/11/20', '2025/09/15',
'2025/11/20', '2025/09/15', '2025/11/20', '2025/09/15',
'2025/11/20'],
    'Subject_Code': ['CS101', 'CS101', 'EC101', 'EC101',
'CS101', 'CS101', 'IT101', 'IT101', 'ME101', 'ME101', 'CS101',
'CS101', 'EC101', 'EC101', 'IT101', 'IT101'],
    'Marks_Obtained': [85, 92, 70, 78, 45, 52, 90, 96, 35, 40,
88, 94, 65, 72, 92, 98],
    'Max_Marks': [100, 100, 100, 100, 100, 100, 100, 100, 100,
100, 100, 100, 100, 100, 100, 100],
    'Enrollment_Status': ['Active', 'Active', 'Active',
'Active', 'Active', 'Active', 'Active', 'Active', 'Dropped',
'Dropped', 'Active', 'Active', 'Active', 'Active', 'Active',
'Active']
}

# Table 2: Student Assessment Log (Term 2) with Mismatched
Column Headers -> Student_Assessments_Term2.csv
# For (d) Merge Mismatched Fields (e.g. Std_ID instead of
Student_ID, Score instead of Marks_Obtained, Dept instead of
Department)

term2_data = {
    'Std_ID': ['S101', 'S101', 'S102', 'S102', 'S103', 'S103',
'S104', 'S104', 'S106', 'S106', 'S107', 'S107', 'S108',
'S108'],
```

```
    'Full_Name': ['Aarav Sharma', 'Aarav Sharma', 'Priya Verma', 'Priya Verma', 'Rohan Gupta', 'Rohan Gupta', 'Sneha Patel', 'Sneha Patel', 'Ananya Singh', 'Ananya Singh', 'Rahul Nair', 'Rahul Nair', 'Kavya Reddy', 'Kavya Reddy'],
    'Dept': ['Computer Science', 'CS', 'ECE', 'Electronics', 'CSE', 'CS', 'IT', 'Information Tech', 'CSE', 'Computer Science', 'ECE', 'Electronics', 'IT', 'IT'],
    'Exam_Type': ['Midterm', 'Final', 'Midterm', 'Final', 'Midterm', 'Final', 'Midterm', 'Final', 'Midterm', 'Final', 'Midterm', 'Final', 'Midterm', 'Final'],
    'Assessment_Date': ['2026/02/10', '2026/04/25', '2026/02/10', '2026/04/25', '2026/02/10', '2026/04/25', '2026/02/10', '2026/04/25', '2026/02/10', '2026/04/25', '2026/02/10', '2026/04/25', '2026/02/10', '2026/04/25'],
    'Course_Code': ['CS201', 'CS201', 'EC201', 'EC201', 'CS201', 'CS201', 'IT201', 'IT201', 'CS201', 'CS201', 'EC201', 'EC201', 'IT201', 'IT201'],
    'Score': [88, 95, 75, 82, 50, 60, 94, 98, 90, 96, 70, 75, 95, 100],
    'Total_Possible': [100, 100, 100, 100, 100, 100, 100, 100, 100, 100, 100, 100, 100, 100],
    'Enrollment_Status': ['Active', 'Active', 'Active', 'Active', 'Active', 'Active', 'Active', 'Active', 'Active', 'Active', 'Active', 'Active', 'Active', 'Active']
}

# Table 3: Course Credit & Grading Benchmark -> Course_Credit_Master.csv

course_data = {
    'Subject_Code': ['CS101', 'CS201', 'EC101', 'EC201', 'IT101', 'IT201', 'ME101'],
    'Subject_Title': ['Data Structures', 'Algorithms', 'Digital Logic', 'Signals & Systems', 'Database Systems', 'Cloud Computing', 'Thermodynamics'],
    'Credit_Hours': [4, 4, 3, 4, 4, 4, 3],
    'Passing_Marks': [40, 40, 40, 40, 40, 40, 40]
}

df_t1 = pd.DataFrame(term1_data)
df_t2 = pd.DataFrame(term2_data)
df_c = pd.DataFrame(course_data)

df_t1.to_csv('Student_Assessments_Term1.csv', index=False)
df_t2.to_csv('Student_Assessments_Term2.csv', index=False)
df_c.to_csv('Course_Credit_Master.csv', index=False)

print("All 3 CSV datasets for Lesson 10 created successfully.")
```

Here is the complete schema design and CSV datasets tailored specifically for Tableau Prep Builder to fulfill all activities of Practical 10 (a through g).

Dataset Architecture and Schema Design

To practice every feature of Tableau Prep (profiling, grouping, merging mismatched union columns, aggregations, calculated filters, and automated flow generation), we use 3 specialized CSV files:

1. Student_Assessments_Term1.csv (Term 1 Assessment Log)
   - Purpose: Data profiling, handling extra spaces, mixed casings, inconsistent date text strings, and spelling variations in departments.
   - Columns: Student_ID, Student_Name, Department, Assessment_Type, Exam_Date, Subject_Code, Marks_Obtained, Max_Marks, Enrollment_Status.
2. Student_Assessments_Term2.csv (Term 2 Assessment Log with Mismatched Schemas)
   - Purpose: Practicing Union and resolving mismatched field names (e.g., Std_ID vs Student_ID, Score vs Marks_Obtained, Course_Code vs Subject_Code).
   - Columns: Std_ID, Full_Name, Dept, Exam_Type, Assessment_Date, Course_Code, Score, Total_Possible, Enrollment_Status.
3. Course_Credit_Master.csv (Subject Metadata & Weightages)
   - Purpose: Prep Joins, weighted GPA calculations, credit hours, and subject categorization.
   - Columns: Subject_Code, Subject_Title, Credit_Hours, Passing_Marks.

Mapping to Practical 10 Objectives

Sub-task	Activity	Practical Execution in Tableau Prep
a	Connect and load data	Connect to text files (Term1, Term2, and Course_Master) in Tableau Prep.
b	Clean and profile data	Use Prep's Profile pane to inspect value distributions, summary cards, and detect data type errors.
c	Group and standardize values	Use Group Values (Pronunciation, Common Characters, or Manual) to unify "Comp Sci", "CS", "CSE" --> "Computer Science".
d	Merge mismatched fields	Union Term 1 and Term 2, then drag and drop mismatched columns (Std_ID onto Student_ID, Score onto Marks_Obtained) to merge them.
e	Perform aggregations in Prep	Add an Aggregate Step to compute student total score, overall average, and exam count grouped by Student_ID & Term.

Sub-task	Activity	Practical Execution in Tableau Prep
f	Apply filters and transformations	Filter out "Dropped" enrollment records, create calculated fields for Percentage, Grade, and Status flags.
g	Create and automate data flows	Build an Output Step to export the cleaned, unified pipeline to a .hyper extract or .csv file.

Raw CSV Data (Create these 3 files on your computer)**.** You can create different csv files using Notepad/text Editor. Copy paste the data in Notepad/text Editor and save the files with .csv extension.

File 1: Student_Assessments_Term1.csv

```
Student_ID,Student_Name,Department,Assessment_Type,Exam_Date,
Subject_Code,Marks_Obtained,Max_Marks,Enrollment_Status
S101,  Aarav Sharma ,Computer
Science,Midterm,2025/09/15,CS101,85,100,Active
S101,Aarav Sharma,Comp
Sci,Final,2025/11/20,CS101,92,100,Active
S102,priya VERMA,Electronics,Midterm,15-09-
2025,EC101,70,100,Active
S102,priya VERMA,ECE,Final,20-11-2025,EC101,78,100,Active
S103,Rohan Gupta,CS,Midterm,2025.09.15,CS101,45,100,Active
S103,Rohan Gupta,CSE,Final,2025.11.20,CS101,52,100,Active
S104,Sneha Patel,Information
Tech,Midterm,2025/09/15,IT101,90,100,Active
S104,Sneha Patel,IT,Final,2025/11/20,IT101,96,100,Active
S105,aditya Rao
,Mechanical,Midterm,2025/09/15,ME101,35,100,Dropped
S105,aditya Rao ,Mech
Engg,Final,2025/11/20,ME101,40,100,Dropped
S106,Ananya Singh,Computer
Science,Midterm,2025/09/15,CS101,88,100,Active
S106,Ananya Singh,CS,Final,2025/11/20,CS101,94,100,Active
S107,Rahul Nair,ECE,Midterm,2025/09/15,EC101,65,100,Active
S107,Rahul Nair,Electronics
Engg,Final,2025/11/20,EC101,72,100,Active
S108,Kavya Reddy,IT,Midterm,2025/09/15,IT101,92,100,Active
S108,Kavya Reddy,Info
Tech,Final,2025/11/20,IT101,98,100,Active
```

File 2: Student_Assessments_Term2.csv

```
Std_ID,Full_Name,Dept,Exam_Type,Assessment_Date,Course_Code,S
core,Total_Possible,Enrollment_Status
S101,Aarav Sharma,Computer
Science,Midterm,2026/02/10,CS201,88,100,Active
S101,Aarav Sharma,CS,Final,2026/04/25,CS201,95,100,Active
S102,Priya Verma,ECE,Midterm,2026/02/10,EC201,75,100,Active
```

```
S102,Priya
Verma,Electronics,Final,2026/04/25,EC201,82,100,Active
S103,Rohan Gupta,CSE,Midterm,2026/02/10,CS201,50,100,Active
S103,Rohan Gupta,CS,Final,2026/04/25,CS201,60,100,Active
S104,Sneha Patel,IT,Midterm,2026/02/10,IT201,94,100,Active
S104,Sneha Patel,Information
Tech,Final,2026/04/25,IT201,98,100,Active
S106,Ananya Singh,CSE,Midterm,2026/02/10,CS201,90,100,Active
S106,Ananya Singh,Computer
Science,Final,2026/04/25,CS201,96,100,Active
S107,Rahul Nair,ECE,Midterm,2026/02/10,EC201,70,100,Active
S107,Rahul
Nair,Electronics,Final,2026/04/25,EC201,75,100,Active
S108,Kavya Reddy,IT,Midterm,2026/02/10,IT201,95,100,Active
S108,Kavya Reddy,IT,Final,2026/04/25,IT201,100,100,Active
```

File 3: Course_Credit_Master.csv

```
Subject_Code,Subject_Title,Credit_Hours,Passing_Marks
CS101,Data Structures,4,40
CS201,Algorithms,4,40
EC101,Digital Logic,3,40
EC201,Signals & Systems,4,40
IT101,Database Systems,4,40
IT201,Cloud Computing,4,40
ME101,Thermodynamics,3,40
```

A: Connect and Load Data into Tableau Prep

B: Clean and Profile Data

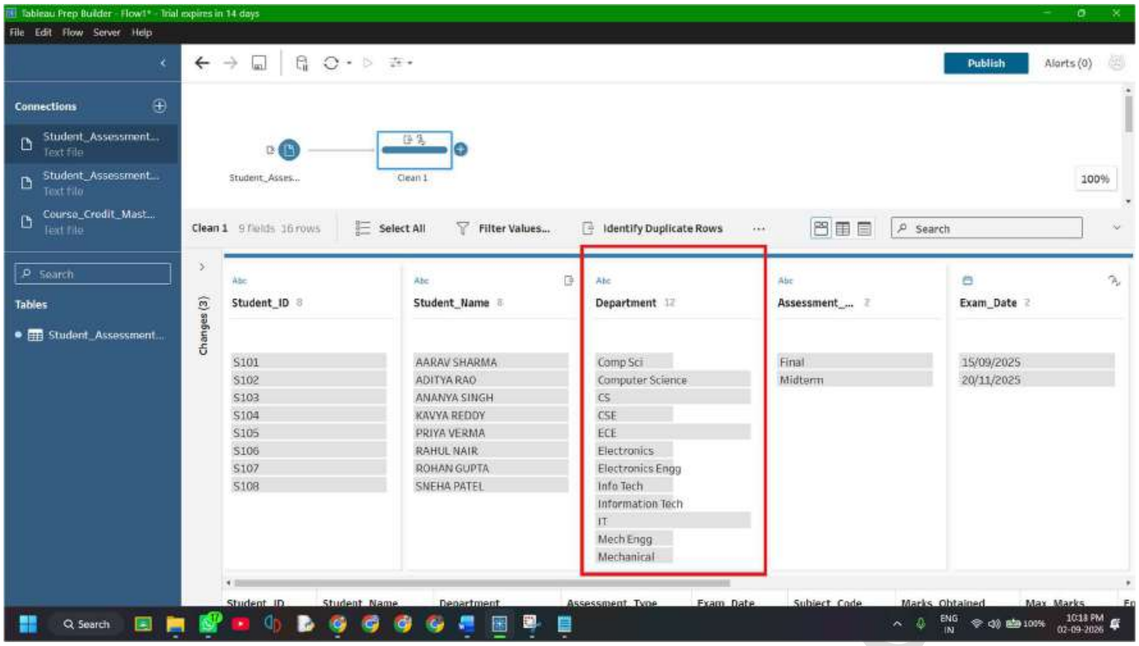
C: Group and Standardize Values

Objective: Standardize spelling variations, abbreviations, and synonyms in categorical columns (such as `Department` and `Dept`) into clean, consistent values using Tableau Prep's built-in grouping algorithms.

In academic datasets, departments are frequently entered under different aliases (e.g., `"Comp Sci"`, `"CS"`, `"CSE"`, and `"Computer Science"`). If left unstandardized, downstream visual analysis will treat each alias as a completely different academic department, splintering the summary metrics.

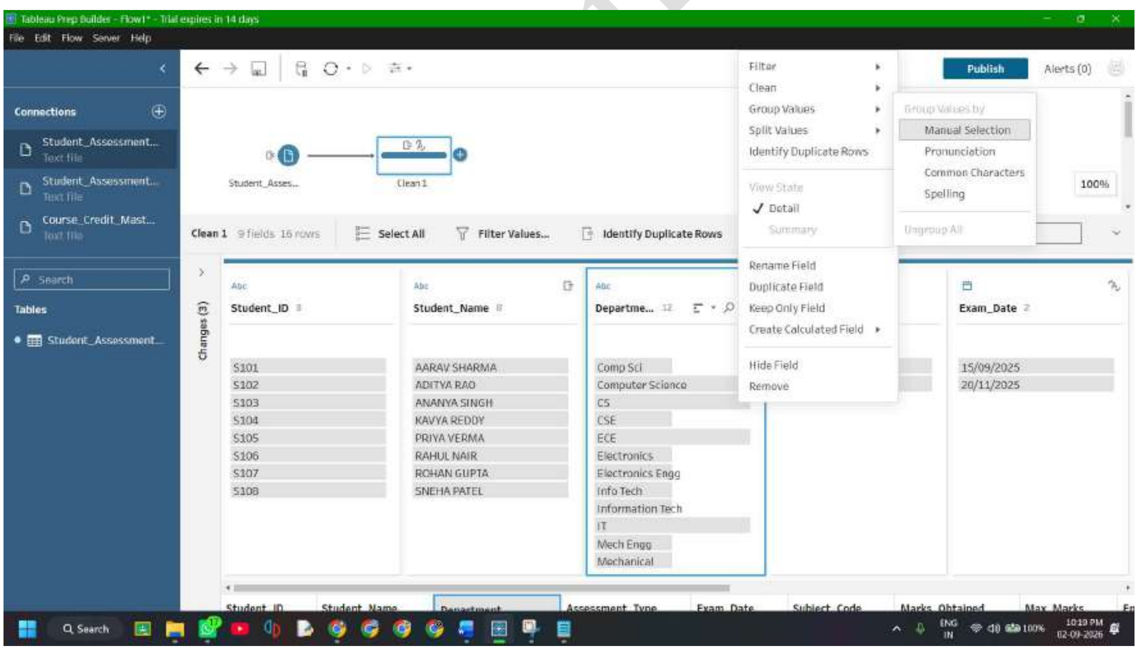
Step 1: Inspect the `Department` Field in Term 1 (`Clean 1`)

1. Click on the **Clean 1** step connected to `Student_Assessments_Term1`.
2. Locate the **`Department`** card in the Profile Pane.
3. Observe that there are 12 distinct entries representing only 4 actual academic departments:
   - Computer Science (`Computer Science`, `Comp Sci`, `CS`, `CSE`)
   - Electronics (`Electronics`, `ECE`, `Electronics Engg`)
   - Information Technology (`Information Tech`, `IT`, `Info Tech`)
   - Mechanical Engineering (`Mechanical`, `Mech Engg`)

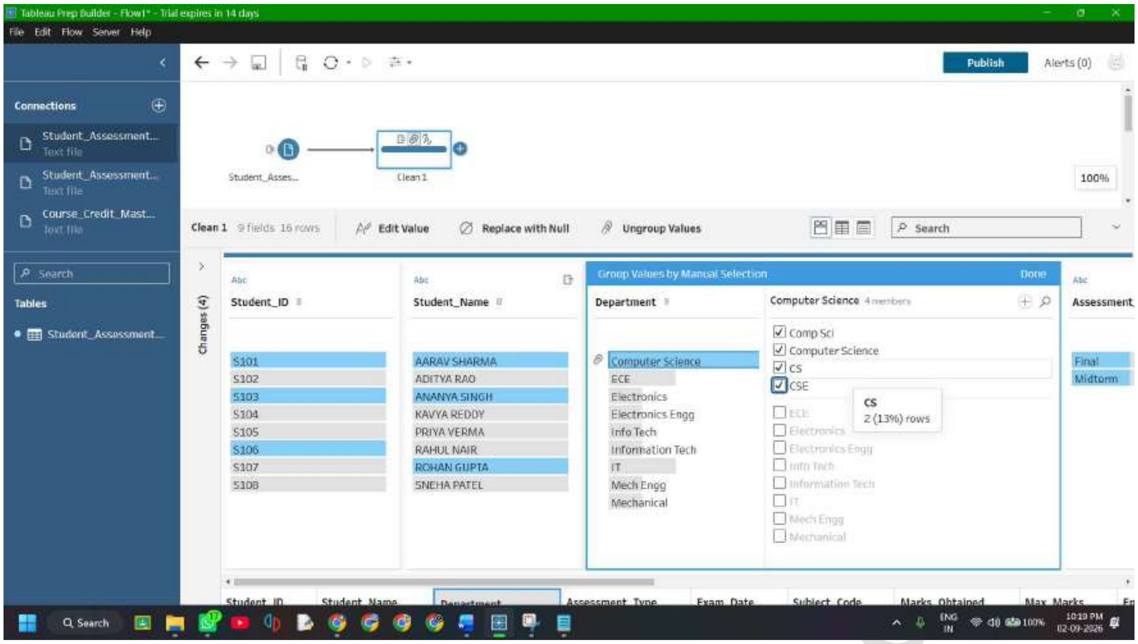


Step 2: Group Values Using Pronunciation / Spelling or Manual Selection

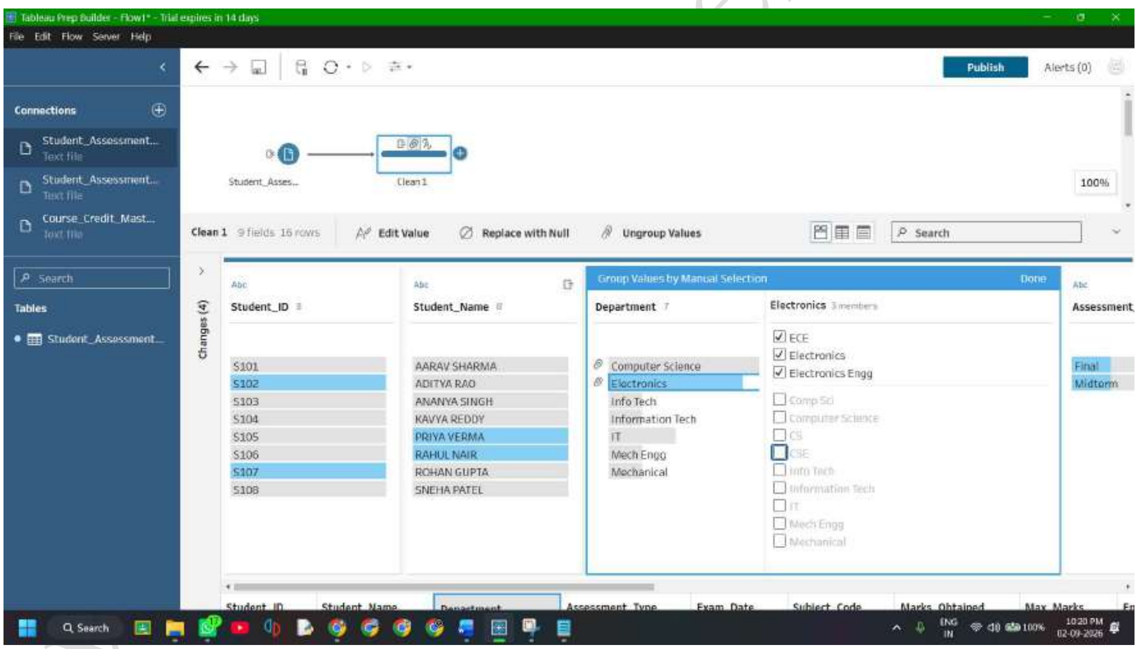
1. On the `Department` card, click the **More options (...)** menu at the top right.
2. Go to **Group Values** -->select **Manual Selection** (or **Pronunciation** / **Common Characters**).



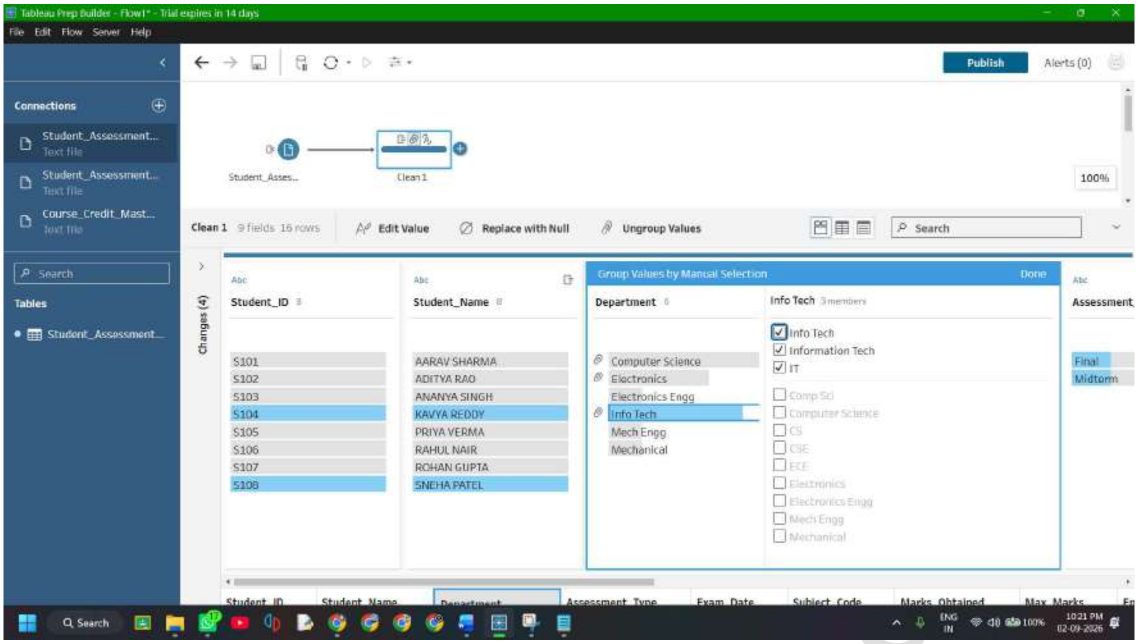
3. In the Grouping editor:
   - Click on `Computer Science` in the left column. In the right-hand panel, check the boxes for `Comp Sci`, `CS`, and `CSE`.



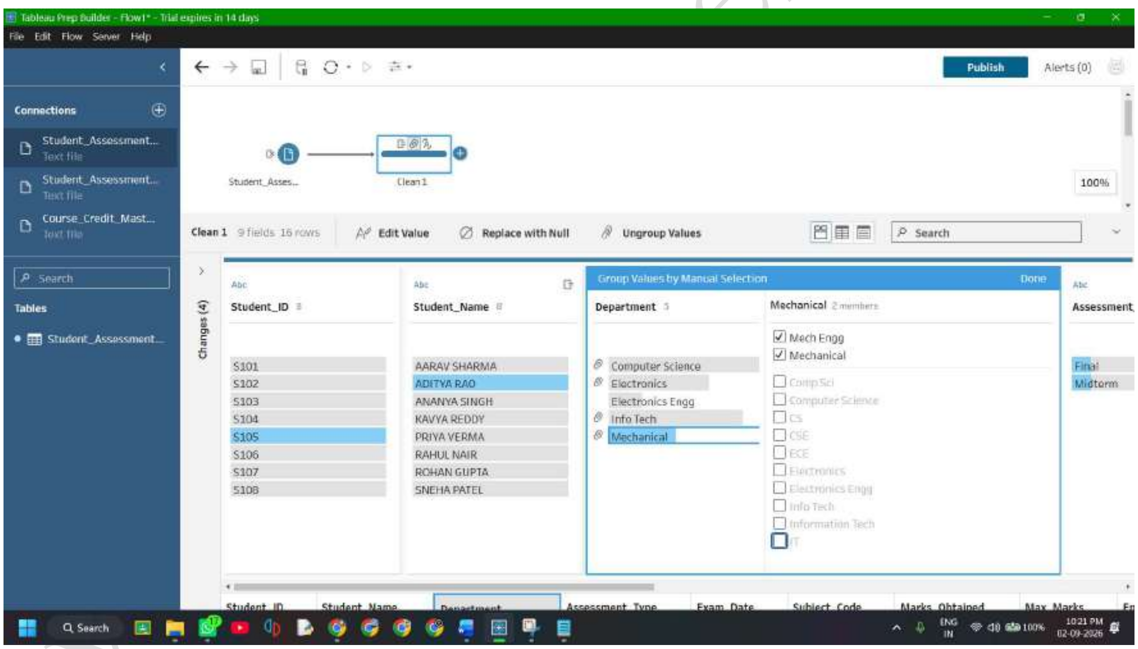
- Click on **`Electronics`** (or rename the master value to `Electronics & Communication`). Check the boxes for **`ECE`** and **`Electronics Engg`**.



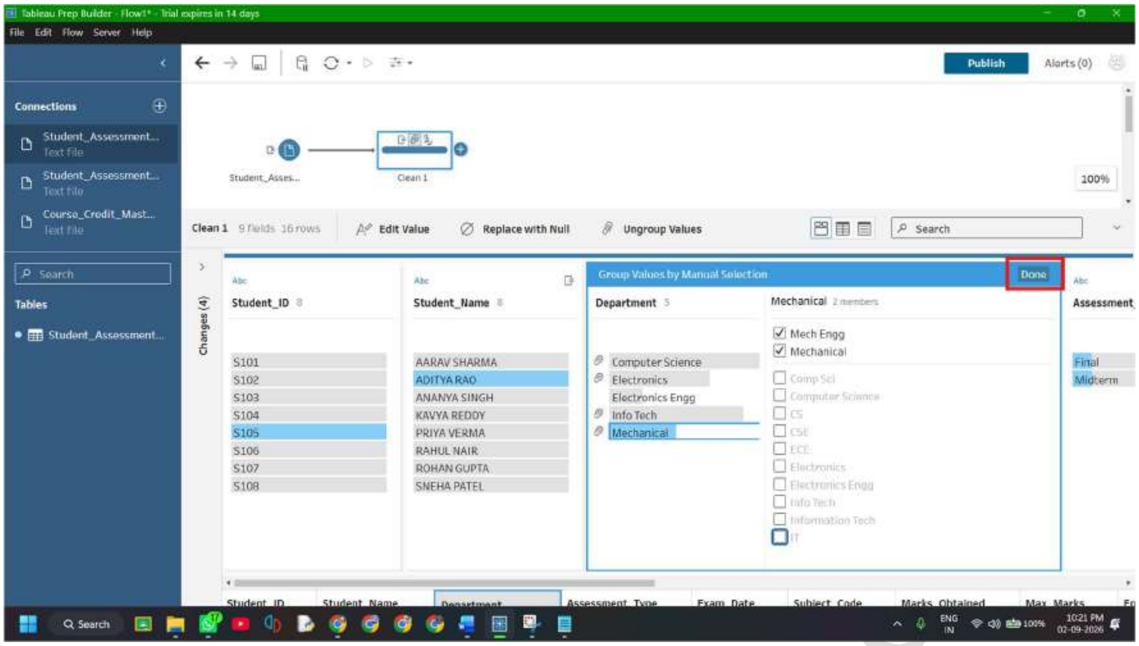
- Click on **`Information Tech`** (or rename to `Information Technology`). Check the boxes for **`IT`** and **`Info Tech`**.



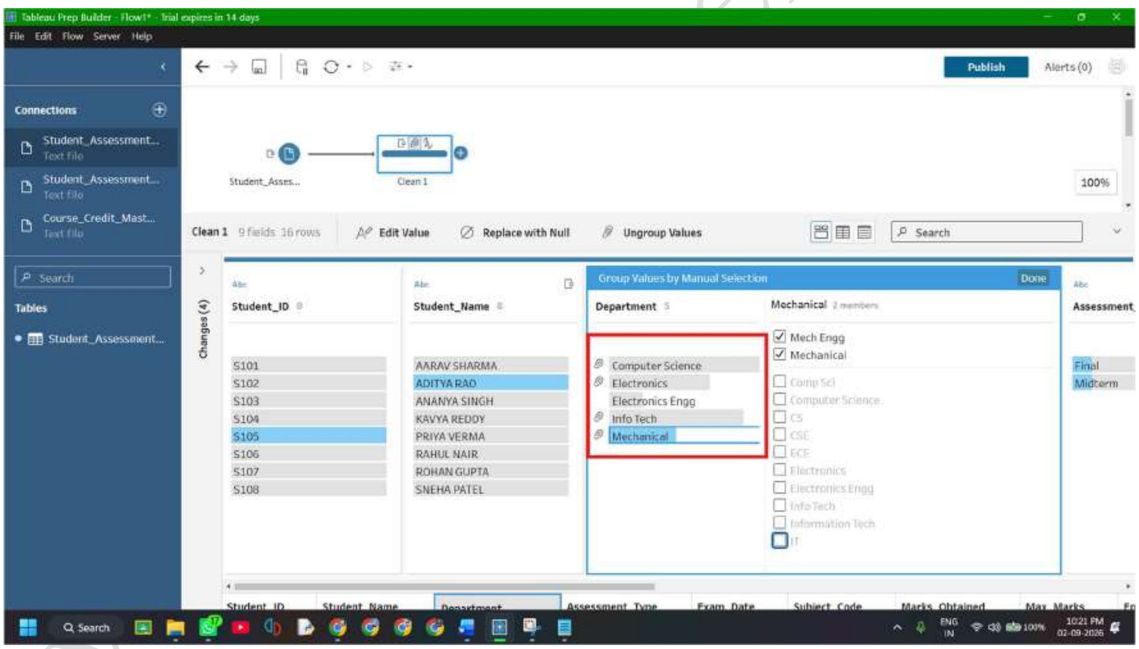
- Click on **`Mechanical`** (or rename to `Mechanical Engineering`). Check the box for **`Mech Engg`**.



4. Click **Done** at the top right of the editor.



5. Notice that the paperclip icon appears next to grouped values, consolidating the 12 messy entries into exactly **4 clean departments**.



Step 3: Repeat Grouping on Term 2 (`Clean 2`)

1. Click on the **Clean 2** step connected to `Student_Assessments_Term2`.
2. Locate the **`Dept`** card.
3. Click **More options (...)** -->**Group Values** -->**Manual Selection**:
   - Group `CS`, `CSE`, and `Computer Science` into **`Computer Science`**.
   - Group `ECE` and `Electronics` into **`Electronics`**.
   - Group `IT` and `Information Tech` into **`Information Tech`**.
4. Click **Done**.

Standardization Mapping Table

Original Input Variations	Target Standardized Value	Department Code
`Computer Science, Comp Sci, CS, CSE`	**Computer Science**	`CS`
`Electronics, ECE, Electronics Engg`	**Electronics**	`EC`
`Information Tech, IT, Info Tech`	**Information Tech**	`IT`
`Mechanical, Mech Engg`	**Mechanical**	`ME`

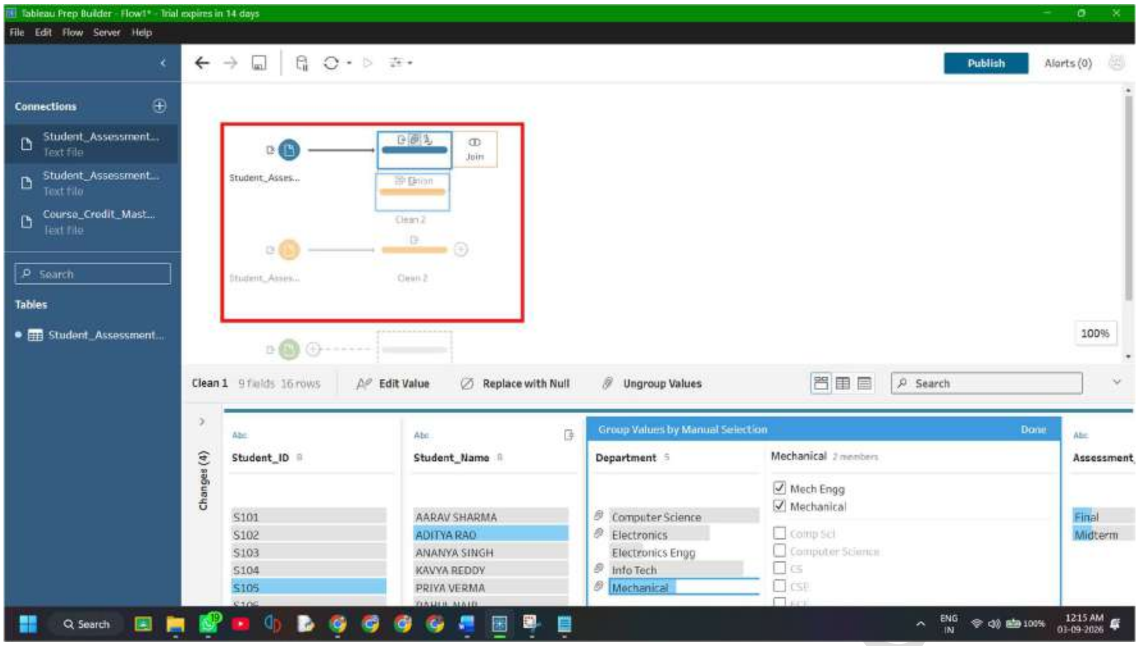
D: Merge Mismatched Fields

Objective: Perform a vertical Union of `Clean 1` (Term 1) and `Clean 2` (Term 2), resolve mismatched column headers (such as `Std_ID` vs `Student_ID`, `Score` vs `Marks_Obtained`), and unify them into a single schema.

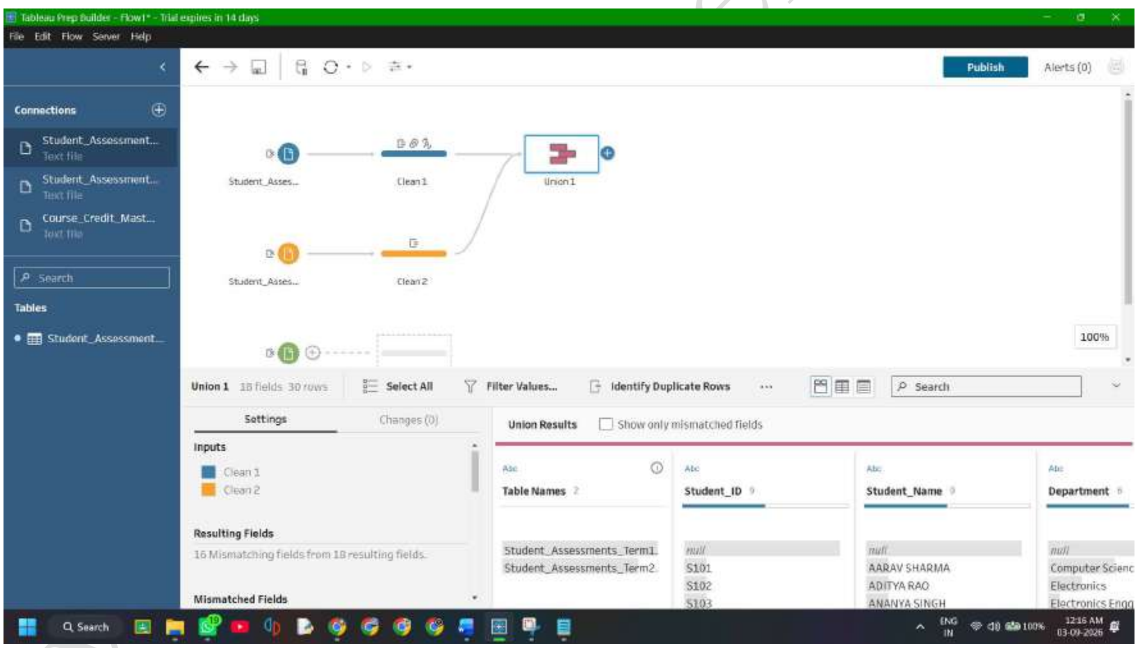
When combining datasets originating from different semesters, departments, or source systems, identical information often resides under dissimilar column names. In Tableau Prep, unmerged fields create sparse datasets with duplicate null columns. Merging mismatched fields consolidates those columns without requiring upstream source file modifications.

Step 1: Create the Union Step on the Canvas

1. On the Flow canvas, click and drag the **`Clean 2`** step icon over to the **`Clean 1`** step icon.

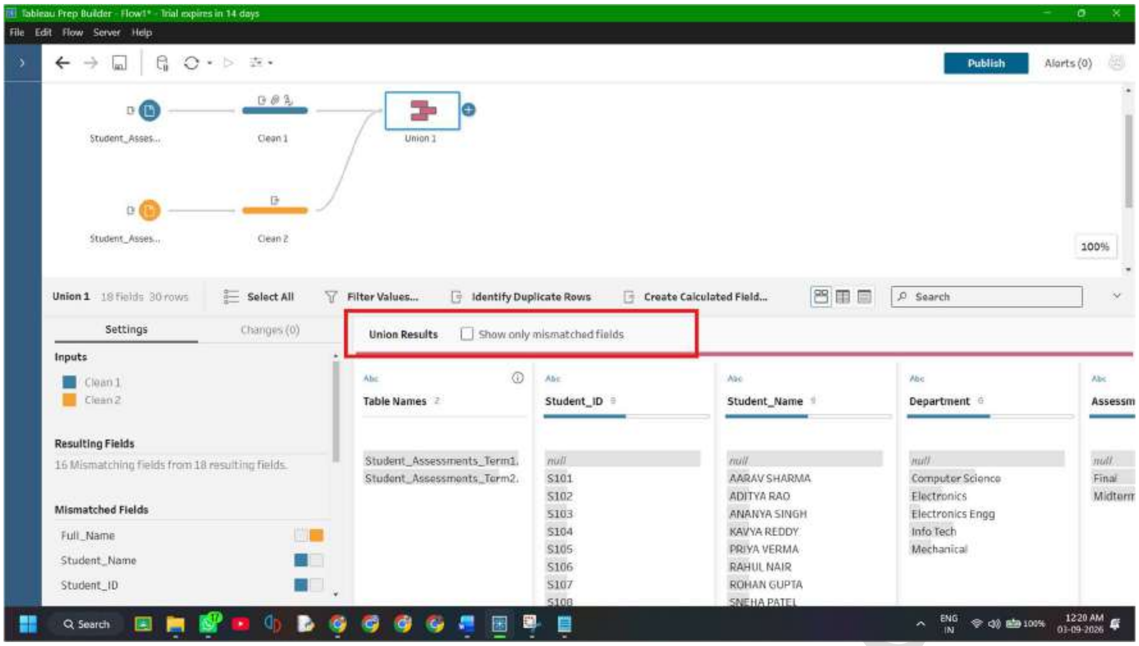


2. When the drop zone appears, hover over **Union** and release the mouse.
3. A new **Union 1** step is created, connecting both transformation branches.



Step 2: Inspect Mismatched Fields in the Union Pane

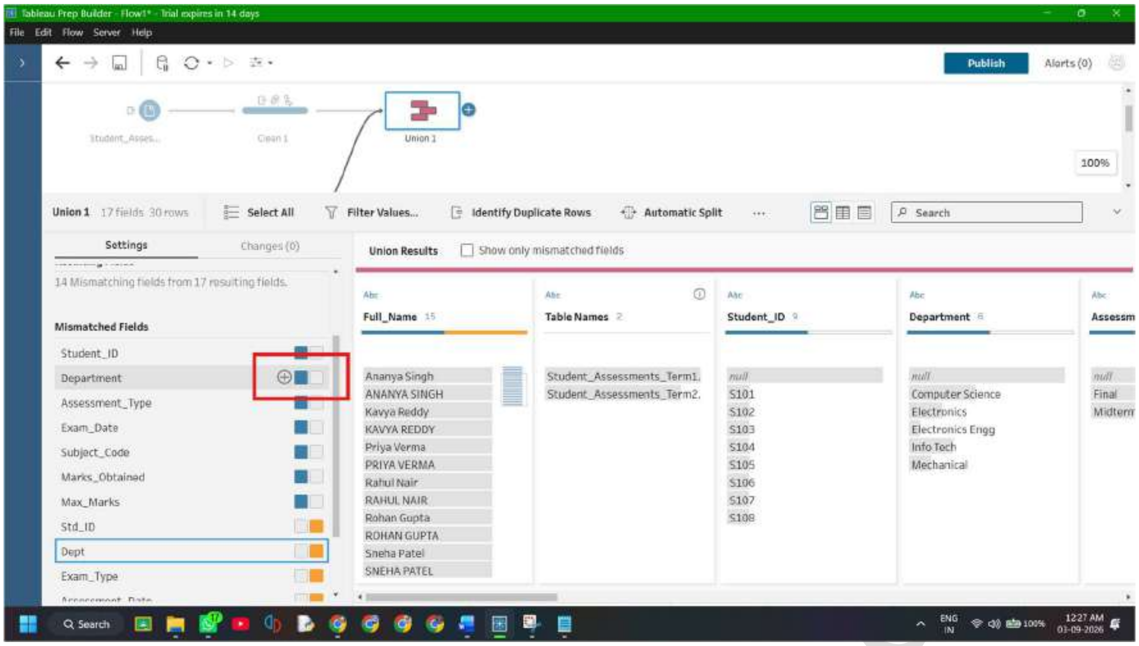
1. Click on the **Union 1** step.
2. In the bottom **Union** summary pane, look at the **Mismatched Fields** section.



3. Notice that because the two files used different naming conventions, Tableau Prep lists non-matching pairs (indicated by yellow and blue bars), such as:
    - `Student_ID` (Term 1) vs `Std_ID` (Term 2)
    - `Student_Name` (Term 1) vs `Full_Name` (Term 2)
    - `Department` (Term 1) vs `Dept` (Term 2)
    - `Assessment_Type` (Term 1) vs `Exam_Type` (Term 2)
    - `Exam_Date` (Term 1) vs `Assessment_Date` (Term 2)
    - `Subject_Code` (Term 1) vs `Course_Code` (Term 2)
    - `Marks_Obtained` (Term 1) vs `Score` (Term 2)
    - `Max_Marks` (Term 1) vs `Total_Possible` (Term 2)

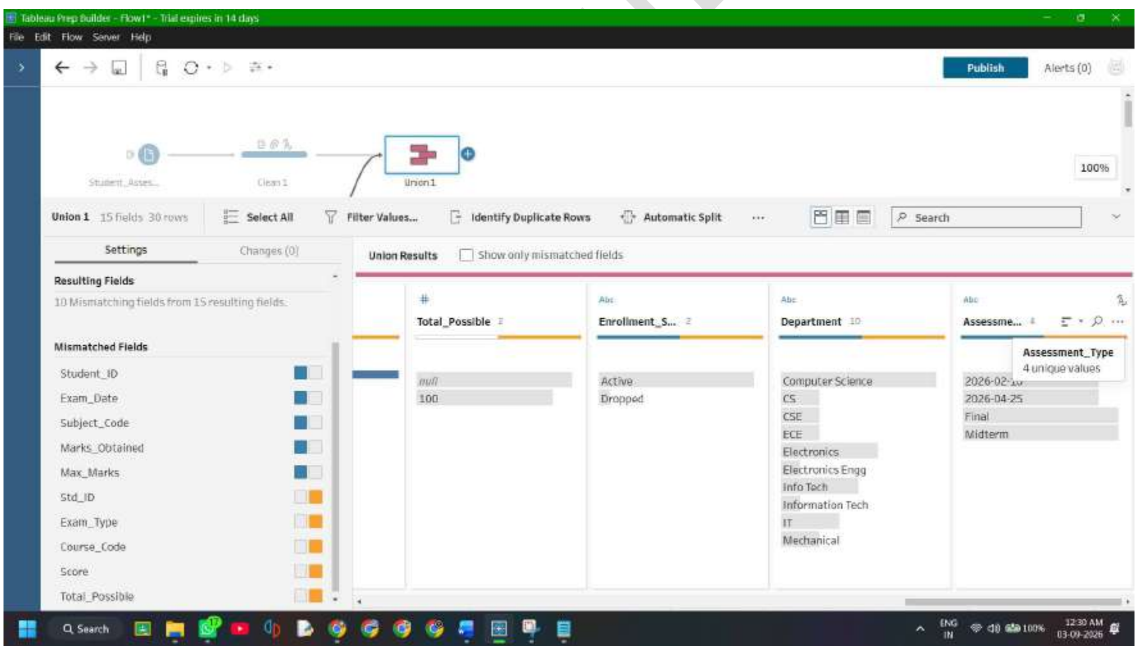
Step 3: Merge the Mismatched Fields

1. In the **Mismatched Fields** list (or the union profile pane):
    - Click and drag **`Std_ID`** directly on top of **`Student_ID`** to merge them into one column.
    - Drag **`Full_Name`** onto **`Student_Name`**.
    - Drag **`Dept`** onto **`Department`**.

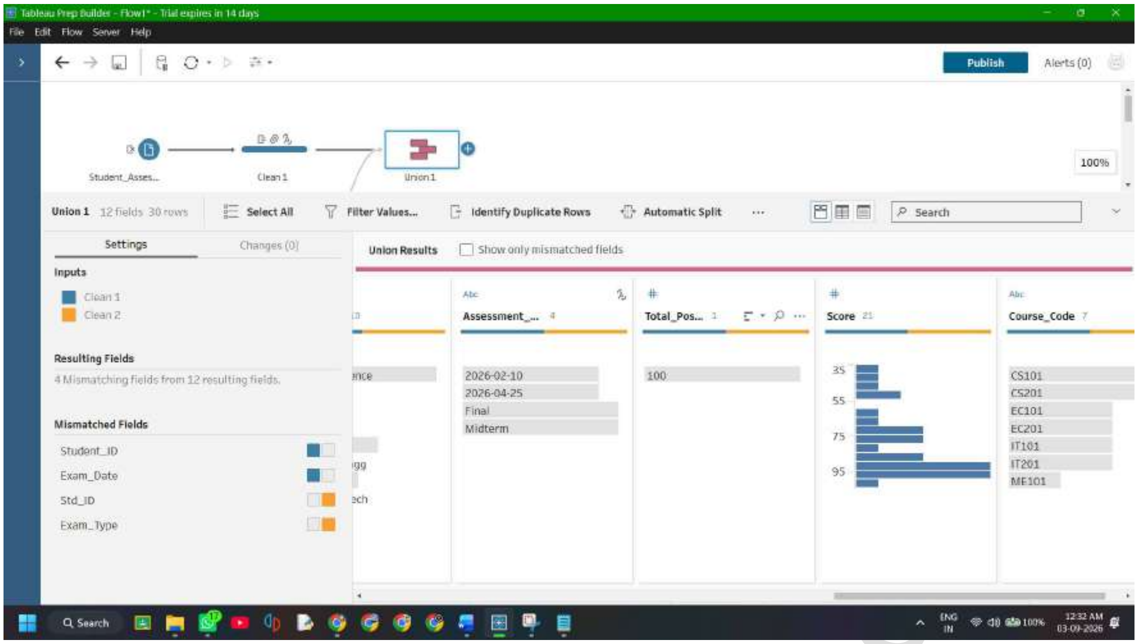


Note: If You cant drag it Observe in the left pane click the dept column the hover near department column's plus icon and add the dept column into it

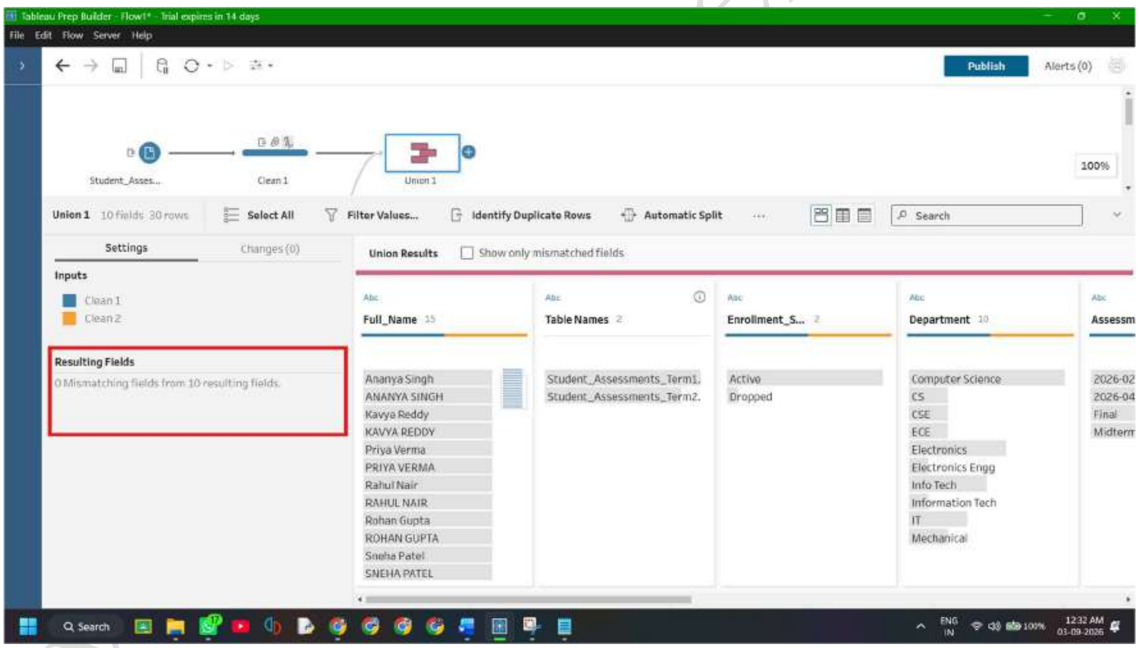
- Drag **`Exam_Type`** onto **`Assessment_Type`**.



- Drag **`Assessment_Date`** onto **`Exam_Date`**.
- Drag **`Course_Code`** onto **`Subject_Code`**.
- Drag **`Score`** onto **`Marks_Obtained`**.
- Drag **`Total_Possible`** onto **`Max_Marks`**.

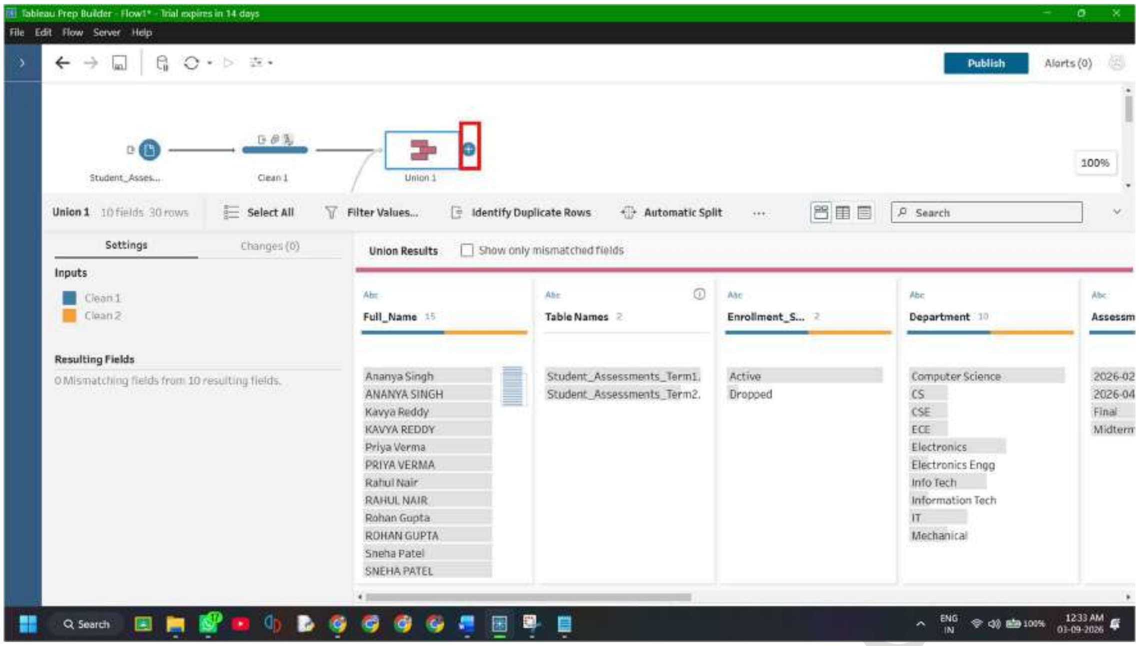


2. Notice that as you merge them, the mismatched count drops to **0**, and each unified column displays both source colors.

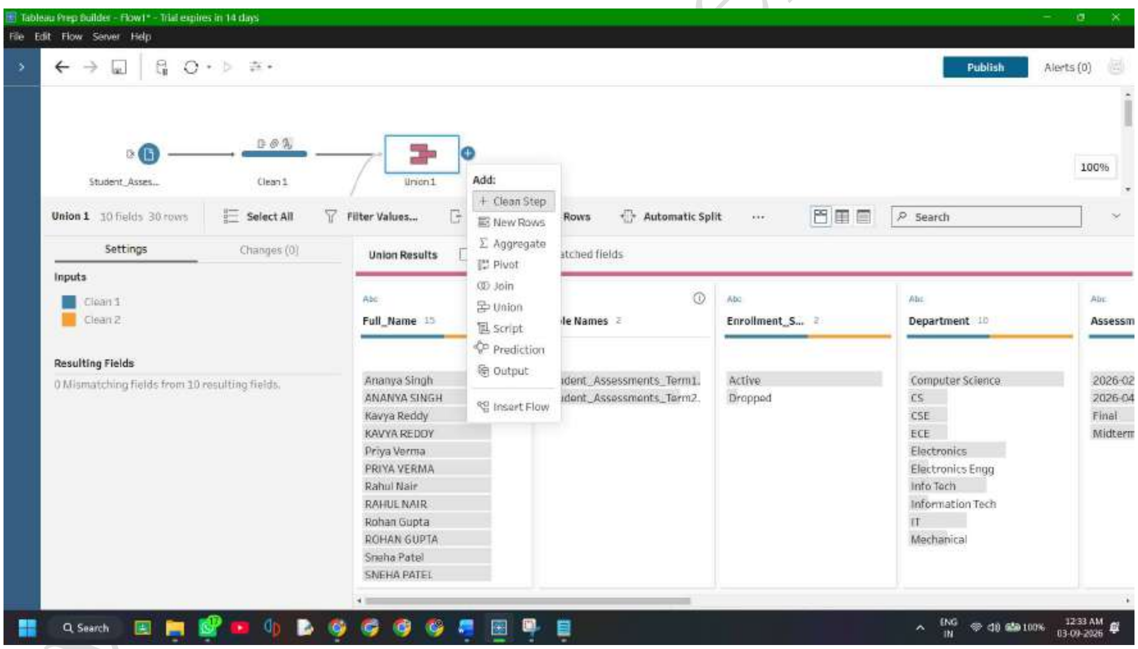


Step 4: Add a Clean Step Post-Union

1. Click the **+** icon next to the **`Union 1`** node.



2. Select **Clean Step** (creates `Clean 3`). *In Screenshot it is showing Clean 4 because I have created clean step for Course_Credit_Master.csv too.*



3. Verify that the combined dataset now contains **30 total rows** across **10 unified columns** (including `Table Names`).